

SQL Data Analysis Report

Project Title - SQL Data Analysis using MySQL

Objective

The objective of this project was to perform SQL-based analysis on an e-commerce sales dataset. The goal was to extract meaningful information using SQL queries and develop practical knowledge of relational database operations.

Tools Used

- MySQL Workbench
 - SQL
 - Microsoft Excel
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Dataset Description

The dataset consists of e-commerce sales transactions containing information about orders, customers, products, quantities, payment methods, referral sources, order status, and total sales.

Tasks Performed

- Imported the cleaned dataset into MySQL.
 - Executed SQL queries for data retrieval.
 - Filtered records using WHERE conditions.
 - Sorted data using ORDER BY.
 - Applied aggregate functions such as COUNT, SUM, AVG, MIN and MAX.
 - Grouped records using GROUP BY.
 - Filtered grouped data using HAVING.
 - Solved business-related analytical questions.
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Key Findings

- Calculated total sales revenue.
- Identified top spending customers.
- Determined best-selling products.
- Compared payment methods based on revenue.
- Calculated average order value.
- Generated monthly sales summaries.
- Analyzed customer referral sources.

Conclusion

This project provided practical experience in writing SQL queries for business analysis. It improved my understanding of relational databases, aggregate functions, filtering techniques, and grouping operations. The project also demonstrated how SQL can be used to convert raw transactional data into meaningful business insights.